Project Proposal

EEE 2124

Section: G

Group No: 2

Group information

Name	ID
Eusa Anjum Himel	0112430162
Faija Ahamed Mim	0112430171
Maryum Binte Hossain	0112430152
Shihab Ahammed	0112330967
Nusrat Ahmed Meem	0112330259

1 Project Title

A Smart, Tracked Wall-Climbing Robot using Dual-BLDC Aerodynamic Adhesion and Active Floor-to-Wall Transition.

2 Project Goal

- Goal 1: Design and implement a tracked locomotion system capable of scaling vertical surfaces utilizing dual A2212 BLDC motors for balanced aerodynamic adhesion.
- Goal 2: Integrate an active mechanical transition arm driven by MG90S servos to physically lift the chassis and facilitate a seamless move from the floor to the wall.
- Goal 3: Develop a multi-sensory feedback loop for real-time orientation tracking (IMU), battery monitoring, and forward obstacle detection during vertical traversal.

3 Project Requirements & Components

- The project utilizes 1 Microcontroller.
- 3 different sensors are used.
- 2 or more passive circuit elements are used (N20 Motors, Resistors).
- One or more active circuit elements are used (LM7805, L298N).

3.1 List of Selected Components

A list of primarily selected components is outlined below. A brief justification behind the selection of each component is provided to explain its role in the hybrid climbing system.

Category	Specific Model	Selection Justification
Microcontroller	ESP32	Selected for its dual-core processing power and built-in Wi-Fi, enabling wireless telemetry and control.
Sensor 1	MPU6050 IMU	Required to measure the robot's tilt and pitch angles to ensure it maintains a safe orientation on the wall.
Sensor 2	HC-SR04 Ultrasonic	Directed forward/upward to detect physical obstacles (like ceilings) in the robot's path to prevent collisions.
Sensor 3	INA219 Voltage/Current	Crucial for safely monitoring the LiPo battery. A voltage drop could cause the BLDC motors to stall, leading to a fall.
Active Element 1	L298N 4WD Motor Driver	A robust motor driver used to control the speed and direction of the N20 micro gear motors driving the tracks.
Active Element 2	FVT LittleBee BLHeli-S ESC	Electronic Speed Controllers are strictly required to translate signals from the ESP32 into high-frequency power for the BLDC motors.
Active Element 3	LM7805 Voltage Regulator	Used to step down the 11.1V from the 3S LiPo battery to a stable 5V for the ESP32 and sensor logic circuits.
Passive Element	Resistor Pack	Includes essential passive components like 10kΩ pull-up resistors for the I2C communication lines (MPU6050 and INA219).
Actuator (Adhesion)	A2212 BLDC Motor (x2)	High-speed brushless motors selected to generate the intense downward aerodynamic thrust required to stick to the wall.
Actuator (Drive)	N20 Micro Gear Motor (x4)	High-torque, lightweight DC motors selected to drive the tracks and pull the robot's weight vertically against gravity.

Category	Specific Model	Selection Justification
Actuator (Vector)	MG90S Servo Motor (x2)	Actuates a mechanical arm to lift the chassis front, enabling a smooth transition from the floor to the wall.
Power Supply	3S 1000mAh LiPo Battery	Provides the necessary high-current discharge rate (C-rating) demanded by the dual BLDC motors and ESCs.
Mechanical 1	5-Inch Propeller (x2)	Sized to fit the chassis while moving enough air to create the vacuum/thrust effect required for wall adhesion.
Mechanical 2	Rubber Tracks (Pair)	Selected over standard wheels to maximize the surface contact area and friction against the wall, preventing slip.
Mechanical 3	Chassis Material & Hardware	The lightweight structural foundation (acrylic, 3D printed PLA, or carbon fiber) along with standoffs and screws holding the system together.

4 Project Management

To ensure the project is completed efficiently and within the trimester deadline, we will utilize an iterative project management strategy. The workflow will be divided equally, with specific responsibilities allocated for mechanical assembly, circuit design, and ESP32 programming. The team will conduct weekly review sessions to monitor progress, address technical bottlenecks immediately, and ensure all hardware and software components align with the predefined budget and timeline.

4.1 Estimated Budget Breakdown

This includes the Estimated Budget Breakdown for the selected component prices.

Component	Unit Cost (Tk)	Quantity	Total Cost (Tk)
ESP32 Microcontroller	600.00	1	600.00
A2212 BLDC Motor	450.00	2	900.00
5-Inch Propeller	50.00	2	100.00
N20 Micro Gear Motor	300.00	4	1200.00
L298N 4WD Motor Driver	200.00	1	200.00
MPU6050 IMU Sensor	200.00	1	200.00
HC-SR04 Ultrasonic Sensor	100.00	1	100.00
ESC FVT LittleBee BLHeli-S	1000.00	2	2000.00
INA219 Voltage/Current Sensor	200.00	1	200.00
LM7805 & Resistor Pack	50.00	1	50.00
3S 1000mAh LiPo Battery	200.00	1	200.00
MG90S Servo Motor	250.00	2	500.00
Chassis Material & Hardware	800.00	1	800.00
Rubber Tracks (Pair)	500.00	1	500.00
Estimated Total			12000 - 15000 Tk

4.2 Project Timeline

This outlines the Project Timeline for the trimester.

Trimester Week	Planned Tasks / Milestones
Week 1-2	Research, component selection, aerodynamic adhesion calculations, and proposal writing.
Week 3-4	Component procurement, custom tracked chassis fabrication, and initial circuit testing.
Week 5-6	Software development, sensor integration, and horizontal track-drive testing.
Week 7-8	Final assembly, dual-BLDC vertical thrust tuning, troubleshooting on walls, and report drafting.

4.3 SWOT Analysis

Strengths

- Excellent traction and stability provided by the continuous rubber tracks compared to standard wheels.
- High aerodynamic adhesion generated by the dual A2212 BLDC motors, allowing the robot to climb porous, textured, and non-magnetic surfaces.
- Real-time telemetry monitoring and wireless control capabilities natively supported by the ESP32 microcontroller's built-in Wi-Fi.

Weaknesses

- High power consumption from running dual BLDC motors will severely limit the operational runtime on a single 1000mAh 3S LiPo battery.
- The heavy payload of multiple motors, ESCs, and a bulky L298N driver makes achieving the perfect thrust-to-weight ratio difficult.

- Significant acoustic noise generated by the dual 5-inch propellers spinning at high RPMs.

Opportunities

- The included MG90S servos provide the perfect opportunity to develop active thrust vectoring, allowing the robot to tilt its propellers and optimize downforce based on the wall's incline.
- The system architecture is scalable for real-world commercial applications, such as inspecting building facades, bridge pillars, or storage tanks.
- The ESP32 platform leaves processing headroom to add a camera module (like an ESP32-CAM) later for visual structural inspections.

Threats

- A sudden voltage sag or power drop (monitored by the INA219) could stall the BLDC motors, resulting in a catastrophic fall and broken hardware.
- Extremely uneven wall textures or deep crevices could break the aerodynamic seal or physically snag the rubber tracks.
- Continuous high-load operation against gravity risks overheating the L298N motor driver or the LittleBee ESCs over time.

4.4 Benchmark Analysis

Feature	Our Project	VertiGo	PerchMobi3	LEeCH Robot
Adhesion Method	Propeller thrust + tracks	Tiltable propellers	Quad fans (thrust reuse)	Suction cups
Surface Compatibility	Smooth + rough + curved	Mostly smooth/irregular walls	Multiple surfaces	Mainly smooth only
Energy Consumption	High	High	Medium (optimized)	Low
Best Use Case	Industrial inspection (pipes, vessels)	Research / agile indoor navigation	Advanced inspection + exploration	Inspection in controlled environments
Mobility Modes	Ground + wall	Ground + wall	Ground + wall + flight	Mostly wall only
Mechanical Complexity	Medium	High	Very High	Medium

While traditional wall-climbing robots typically rely on pneumatic suction cups, those implementations are heavily restricted to perfectly smooth surfaces like glass. By utilizing a tracked chassis paired with aerodynamic thrust via dual high-speed BLDC propellers, our proposed system improves upon these existing implementations by being surface-agnostic. It can climb textured, porous surfaces such as brick or concrete while maintaining a larger grip area than wheeled robots. Furthermore, the integration of MG90S servos provides a mechanical advantage for directional control or sensor sweeping that static earlier models lack.

5 References

1] https://www.researchgate.net/publication/335497260_EJBot-II_an_optimized_skid-steering_propeller-type_climbing_robot_with_transition_mechanism

2] https://www.youtube.com/watch?v=xXKKpNN5MZE

3] https://arxiv.org/pdf/2509.12620

4] https://www.wevolver.com/specs/vertigo

5] https://www.therobotreport.com/wall-climbing-robot-inspired-by-the-soft-body-of-a-leech/

6] https://drive.google.com/file/d/1vf5df7R-0sVrojpRzOrvqFdXTq91Rmjo/view?usp=drive_link